Gourab Roy

Erasmus Mundus IPCVAI Master's Student

gourabroysec553@gmail.com | +33 7 51 06 51 37 | Bordeaux, France

Portfolio | GitHub | LinkedIn | Google Scholar

Available from 1 February 2027 for a 5–6 month on-site internship.

Profile

Master's student in AI for image processing and computer vision, with two years of research-assistant experience and co-authorship of 12 peer-reviewed papers. Hands-on work in explainable AI for text and vision–language transformers (BERT, CLIP), medical image analysis (CT/MRI), object detection, 3D vision, and multimodal retrieval. I build reproducible PyTorch pipelines covering data curation, training, evaluation, error analysis, and serving (FastAPI, Docker). I am seeking an internship in a research lab or industrial R&D team to develop, evaluate, and explain vision models.

Education

Erasmus Mundus Joint Master's Degree – IPCVAI **2025 – 2027**

Artificial Intelligence for Image Processing and Computer Vision *France, Spain, Hungary*

- **Université de Bordeaux, France** (Sep 2026 – Present): Explainable AI for Images, Advanced Computer Vision, Acquisition, Reconstruction and Medical Imaging, AI and Sustainability.
- **Universidad Autónoma de Madrid, Spain** (Feb – Jun 2026): Deep Learning for Computer Vision, Deep Learning for Visual Signal Processing, HPC for Deep Learning, Vision for Multiple and Moving Cameras.
- **Pázmány Péter Catholic University, Hungary** (Sep 2025 – Jan 2026): Image Processing, Fundamentals of Deep Learning, Data Mining and ML, Multimodal Sensor Fusion, Biomedical Signal Processing.

B.Sc. in Computer Science and Engineering **2018 – 2023**

Sylhet Engineering College; CGPA: 3.56/4.00 *Sylhet, Bangladesh*

- Thesis: *CerebralNet*, a deep-learning system that detects major abnormalities in brain CT and MRI scans using segmentation and classification.

Research Experience

Tutored Research and Development Project (TRDP), Master's Course **Sep 2025 – Present**

Explainability of text and vision–language transformers *Hungary / Spain / France*

- Supervised by Prof. Jenny Benois-Pineau (Univ. Bordeaux) and Prof. Marcos Escudero-Viñolo (UAM).
- Exploring how attention-based explanations (RFEM, Ayyar et al., 2025) behave beyond Vision Transformers; found that attention-sink tokens can hide the words that drive BERT and CLIP predictions.
- Comparing them with perturbation-based Partition SHAP to see which better reflects the model's reasoning.

Research Assistant **Nov 2023 – Oct 2025**

Sylhet Engineering College, Dept. of CSE; supervisor: Prof. Md. Abu Naser Mojumder *Sylhet, Bangladesh*

- Designed models, ran experiments and evaluations, and prepared manuscripts in medical AI, document AI, agricultural vision, NLP, and structural computer vision; this work led to 12 peer-reviewed publications.
- Contributed to new image datasets (BanglaFont45, Bangladeshi rice varieties) and to LIME-based explainability analyses of CNN classifiers.
- Mentored undergraduate students on final-year theses, experimental design, and scientific writing.

Selected Publications

Journal Articles

[J1] T. R. Ornob, **G. Roy**, and E. Hassan, "CovidExpert: A Triplet Siamese Neural Network Framework for the Detection of COVID-19," *Informatics in Medicine Unlocked*, vol. 37, Art. no. 101156, 2023, doi: 10.1016/j.imu.2022.101156.

[J2] S. A.-A. Shuvo, K. Rahman, A. A. N. Chowdhury, **G. Roy**, N. Nahid, M. Hasan, M. A. N. Mojumder, M. J. A. Soeb, and M. F. Jubayer, "Image-Based Classification of Bangladeshi Rice Varieties Using Deep Convolutional Neural Networks," *Agricultural Products Processing and Storage*, vol. 2, Art. no. 26, 2026, doi: 10.1007/s44462-026-00061-9.

[J3] K. S. Islam, **G. Roy**, N. Nahid, S. Y. Ripti, M. A. N. Mojumder, M. J. A. Soeb, and M. F. Jubayer, "Advancing Bangla Typography: Machine Learning and Transfer Learning Based Font Detection and Classification Using the BanglaFont45 Dataset," *Journal of Computer Sciences Institute*, vol. 35, pp. 166–174, 2025, doi: 10.35784/jcsi.7120.

[J4] T. Islam, S. K. H. Omur, N. Nahid, L. Chowdhury, **G. Roy**, M. A. N. Mojumder, M. J. A. Soeb, and M. F. Jubayer, "Optical Character Recognition for Ancient Scripts: A Case Study on Syloti Nagri Using Deep Learning Models," *Journal of Computer Sciences Institute*, vol. 34, pp. 98–107, 2025, doi: 10.35784/jcsi.6785.

Conference Papers

[C1] A. R. Samad, N. Rahman, M. Z. Islam, M. R. Mahin, **G. Roy**, and M. A. N. Mojumder, "PlantXNet: A CNN-Based Explainable AI Framework for Multi-Plant Leaf Disease Classification Using LIME," in *Proc. IEEE QPAIN*, 2026, doi: 10.1109/QPAIN69676.2026.11545914.

[C2] A. R. Samad, S. Mahmud, M. U. Sayem, M. R. Mahin, S. Hussen, and **G. Roy**, "MaizeXNet: A CNN-Based Explainable AI Framework for Maize Pest and Disease Classification," in *Proc. IEEE QPAIN*, 2026, doi: 10.1109/QPAIN69676.2026.11546210.

[C3] A. R. Samad, M. R. Mahin, P. Routh, S. Hussen, A. K. M. A. Musa, and **G. Roy**, "PoultryXNet: An Explainable Deep Learning Framework for Poultry Disease Classification From Fecal Images," in *Proc. IEEE QPAIN*, 2026, doi: 10.1109/QPAIN69676.2026.11545990.

[C4] A. R. Samad, M. R. Mahin, M. R. Hasan, J. R. Joy, Fahmiduzzaman, **G. Roy**, and M. A. N. Mojumder, "DateFNet: From Pixels to Plates – Attention Based Multi-Stream CNN for Date Fruit Classification," in *Proc. IEEE ICCIT*, 2025, doi: 10.1109/ICCIT68739.2025.11491249.

[C5] M. U. Gazi, N. I. Rupom, G. B. Durjoy, **G. Roy**, N. Nahid, M. M. Hossain, and M. T. Hasan, "Enhancing Concrete Performance: Machine Learning Models for Predicting and Classifying Tensile Strength," in *Proc. IEEE ECCE*, pp. 1–6, 2025, doi: 10.1109/ECCE64574.2025.11013991.

[C6] M. U. Gazi, G. B. Durjoy, **G. Roy**, N. Nahid, M. M. Hossain, and G. Saha, "CrackClassification4: Classifying Four Types of Cracks in Structural Surfaces Using CNNs," in *Proc. IEEE ICCIT*, pp. 2506–2511, 2024, doi: 10.1109/ICCIT64611.2024.11022574.

[C7] M. M. Hossain, N. Nahid, **G. Roy**, M. A. N. Mojumder, and M. S. Rahman, "Optimizing Multilingual Summarization: Advanced Corpus Creation and Filtering Approach," in *Proc. IEEE ICCIT*, pp. 2482–2487, 2024, doi: 10.1109/ICCIT64611.2024.11022549.

Selected Projects

BanglaCaption40K: Bengali Image–Text Retrieval Benchmark *2026*

– Built a benchmark with 217,526 captions and 43,868 images. Lightweight CLIP fine-tuning (0.33% of the parameters) raised image-to-text R@1 from 6.17 to 17.64 and text-to-image R@1 from 1.59 to 11.26; analysed behaviour with token ablation, patch perturbation, and error analysis.

SiteGuard AI: PPE Detection for Construction Safety *Sep 2026 – Present*

– Leading the ML part of a team project: training and comparing YOLO26n/s and YOLOv8n detectors for five-class PPE detection with worker-to-PPE matching, served through FastAPI and Docker.

Calibrated Multi-View 3D Reconstruction *2026*

– MATLAB pipeline: calibration, RANSAC matching, epipolar geometry, pose estimation, sparse reconstruction, and bundle adjustment; 4.47 px reprojection error over 672 multi-view tracks.

Cardiac Autonomic Assessment from PPG-Derived HRV *Dec 2025*

– Built a Python pipeline (Butterworth filtering, adaptive peak detection, IBI cleaning, time- and frequency-domain HRV) for 10 participants under three sound conditions, with Friedman and Wilcoxon tests and subject-level analysis.

Teaching and Academic Service

Adjunct Lecturer **Jan 2024 – Aug 2025**
Sylhet Engineering College, Department of CSE *Sylhet, Bangladesh*

– Taught Introduction to Data Science, Distributed Computing, Computer Architecture, and Microprocessors and Microcontrollers (theory and lab sessions); designed assessments and co-supervised final-year theses.

– **Peer reviewer** for *Informatics in Medicine Unlocked* (Elsevier).

Technical Skills

Programming: Python, C, C++, MATLAB, Java, JavaScript/TypeScript, SQL
Deep Learning: PyTorch, TensorFlow, scikit-learn; CNNs, ViT, CLIP, BERT, YOLO; transfer and few-shot learning
Vision and XAI: OpenCV; classification, segmentation, detection, 3D vision; attention rollout, SHAP, LIME
Scientific Computing: NumPy, SciPy, Pandas, Matplotlib, Jupyter; signal processing and statistical testing
Tools: Git, Linux, Docker, FastAPI, pytest, LaTeX; reproducible experiments, technical and academic writing

Additional Information

– **Certifications:** Machine Learning Specialization and Data Structures (Coursera); 350+ problems solved on HackerRank, Codeforces, and UVA. **Languages:** English and Bengali (native). **References:** available upon request.